Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/25/2007
PI Name:	Pavalko , Fredrick M Ph.D.		
Project Title:	Regulation Of Osteoblast Apoptosis By Fluid Shear Stress		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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City:	Indianapolis	**State:**	IN
Zip Code:	46202-5120	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1606		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Bone mass is increased in response to mechanical loading, while chronic unloading of bone, such as occurs during prolonged bed rest and in microgravity during space flight, results in atrophy of bone. Exposure of osteoblasts and osteocytes to fluid shear stress (FSS), induced by the movement of interstitial fluid within the spaces inside bone, has been hypothesized to play a role in the response of bone to mechanical loading. This hypothesis is supported by in vitro studies demonstrating that bone-forming osteoblasts and osteocytes are stimulated by FSS. Among the responses of bone cells to FSS are the release of autocrine/paracrine signaling factors including prostaglandins and nitric oxide (NO), both of which are known to play a role in bone growth and remodeling. In this project we are seeking to gain a better understanding of the cellular and molecular events that mediate the response of bone to loading. We are examining the effects of steady and pulsatile FSS on growth and survival of osteoblasts. Our results to date suggest that FSS promotes osteoblast survival in vitro. These results raise the possibility that mechanical loading may promote bone formation by increasing the lifespan of osteoblasts in bone through inhibition of osteoblast apoptosis. Our primary objective is to test		

	the hypothesis that FSS plays a role in regulating osteoblast survival using an in vitro cell culture model. Identify cellular mechanisms that regulate apoptosis in bone cells. Inhibition of osteoblast apoptosis could improve bone growth and remodelling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research impacts overall bone health by trying to understand the fundamental cellular and molecular mechanisms that regulate bone remodeling.
Task Progress:	During the past year we have continued to focus on trying to understand the cellular mechanisms that mediate the anti-apoptotic effects of fluid shear stress. We have continued to focus our efforts on understanding the effects of steady and pulsatile fluid shear stress on osteoblasts. We published three manuscripts in the past year including: Charoonpatrapong, K., Shah, JR, Robling, A.G., Alvarez1, M., Clapp, D.W., Chen, S., Kopp, R.P, Pavalko, F.M., Yu, J., and Bidwell, J.P. (2006) HMGB1 expression and release by bone cells, Journal of Cellular Physiology, 207:480-490. Triplett, J.W. and Pavalko, F.M. (2006) Disruption of alpha-actinin-integrin interactions at focal adhesions renders osteoblasts susceptible to apoptosis. American Journal of Physiology: Cell Physiology, 291:909-921. Ponik, S.M., Triplett, J.W. and Pavalko, F.M. (2007) Osteoblasts and osteocytes respond differently to oscillatory and unidirectional fluid flow profiles. Journal of Cellular Biochemistry, 2007 Feb 15;100(3):794-807. Ponik, S.M., Triplett, J.W. and Pavalko, F.M. Osteoblasts and Osteocytes Respond Differently to Oscillatory and Unidirectional Fluid Flow Profiles, . Manuscript submitted to Bone, December 2005). We have also published a study describing the beneficial effects of etoposide on adenoviral-mediated protein expression in osteoblasts, a method that has improved our ability to express exogneous proteins in osteoblasts, a procedure that has been notoriously difficult (Triplett, J.W., Herring, B.P. and Pavalko, F.M. (2005) Adenoviral transgene expression enhanced by co-treatment with etoposide in cultured cells. Biotechniques, 39:826-832). We have continued to investigate the mechanisms of action through which IGF-1 and fluid shear stress act synergistically to enhance the ability of both fluid shear stress and IGF-1 to inhibit apoptosis in bone cells. We have continued to focus on understanding the cellular mechanisms responsible for this “sensitization” of IGF-1 receptors by fluid shear stress. To that end, we are revising a manuscript describing the effect of fluid shear stress on signaling via the IGF-1 receptor and the beneficial effects of shear on IGF-1-mediated inhibition of apoptosis in osteoblasts (Pavalko, F.M., Gerard-O’Riley, R., Tekulve, K., and Norvell, S.M. Fluid Shear Stress Enhances IGF-1 Receptor Signaling in Osteoblasts as a Mechanism to Inhibit Apoptosis. Manuscript submitted). Interestingly, sensitization of the IGF-1 receptor by fluid shear stress is blocked when PKC-zeta activity is inhibited in osteoblasts. This result suggests that regulation of PKC-zeta activity by fluid shear stress may play a role in regulating IGF-1-mediated pro-life signals in bone cells. Together, our results suggest that fluid shear stress may, in part, regulate apoptosis in osteoblasts via a PKC-zeta -dependent sensitization of IGF-1 receptor signaling. We have also been investigating the effects of altered cell-substrate attachments on basal levels of apoptosis in osteoblasts, as well as on the susceptibility of bone cells to apoptotic stimuli. This work has led to two published manuscripts (Ponik, S.M. and Pavalko, F.M. (2004) Formation of Focal Adhesions on Fibronectin Promotes Fluid Shear Stress Induction of COX-2 and PGE2 Release in MC3T3-E1 Osteoblasts. J. Appl. Physiol., 97:135-142; Triplett, J.W. and Pavalko, F.M. (2006) Disruption of alpha-actinin-integrin interactions at focal adhesions renders osteoblasts susceptible to apoptosis.. American Journal of Physiology: Cell Physiology, 291:909-921. These studies should provide information on how bone health is maintained at the cellular level. We will continue to address the specific aims as originally described. In addition, we will continue to explore the role of shear stress on growth factor receptors and examine the possibility that fluid shear stress regulates apoptosis through modulation of PKC-zeta activity, perhaps through a mechanism that involves activation of MAPK signaling pathways. No major changes for the original aims are forseen.
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Charoonpatrapong-Panyayong K, Shah R, Yang J, Alvarez M, Pavalko FM, Gerard-O'riley R, Robling AG, Templeton E, Bidwell JP. "Nmp4/CIZ contributes to fluid shear stress induced MMP-13 gene induction in osteoblasts." J Cell Biochem. 2007 Apr 23; [Epub ahead of print] PMID: 17455210 , May-2007
Articles in Peer-reviewed Journals	Ponik SM, Triplett JW, Pavalko FM. "Osteoblasts and osteocytes respond differently to oscillatory and unidirectional fluid flow profiles." J Cell Biochem. 2007 Feb 15;100(3):794-807. PMID: 17031855 , Feb-2007
Articles in Peer-reviewed Journals	Triplett JW, Pavalko FM. "Disruption of alpha-actinin-integrin interactions at focal adhesions renders osteoblasts susceptible to apoptosis." Am J Physiol Cell Physiol. 2006 Nov;291(5):C909-21. PMID: 16807302 , Nov-2006